

DHRITI MITTAL

[LinkedIn](#) • [GitHub](#) • dhritimittal191205@gmail.com • +91 9975717082

Full-Stack & Applied AI Engineer focused on frontend design and development, with experience shipping polished, production-grade web applications. Also builds applied AI/ML systems - RAG pipelines, distributed ML infrastructure, and deep-learning models - engineering the guardrails that catch what AI outputs get wrong.

EDUCATION

B.Tech in Computer Science and Engineering (Core) | *Vellore Institute of Technology* 2023 – Present
CGPA: 8.29

Class 12th (CBSE) | *Viraj Shri Ram Centennial School, Maharashtra* 2023
Percentage: 87.2%

Class 10th (CBSE) | *Viraj Shri Ram Centennial School, Maharashtra* 2021
Percentage: 99.6%

SKILLS

Programming: Python, TypeScript/JavaScript, C++

AI/ML: PyTorch, Deep Learning, Computer Vision, RAG, Prompt Engineering, Multi-Agent Systems

Full Stack: React, Next.js, FastAPI, REST APIs, Postman, Git/GitHub, Figma

EXPERIENCE

Technical Summer Intern | *LearningShala* ([Project Link ↗](#)) June - July 2026

- Analyzed and redesigned course discovery, admissions, and counselling user journeys in React/Next.js/Tailwind, iterating directly with the founder to reduce user friction and improve conversion.
- Replaced hardcoded styling with a semantic design-token system in Tailwind v4, enabling scalable multi-university theming and reducing repeated page-level styling work.
- Migrated 250+ course pages from static data to an API-driven CMS pipeline, restructuring nested data and resolving Vercel build/configuration failures to restore reliable production delivery.
- Tested and validated the REST API endpoints powering the CMS pipeline using Postman, catching integration issues before they reached production.

PROJECTS

RepoSearch AI: Autonomous Codebase Analyst | *LangChain, LangGraph, FastAPI, ChromaDB, RAG* ([Link ↗](#))

- Built an autonomous codebase-analysis system that routes complex developer queries across 6 task-specialized agents for bug hunting, refactoring, and repository analysis.
- Designed a hybrid RAG pipeline combining BM25, MMR, cross-encoder reranking, and language-aware code chunking to improve retrieval quality across multi-language codebases.
- Built a FastAPI/Next.js application with real-time SSE streaming and privacy-preserving offline analysis via Ollama.

PINN-Based Human Motion Prediction | *PyTorch, Computer Vision, PINNs* | *Ongoing*

- Architecting a physics-informed neural network (PINN) that combines vision-based 3D human joint tracking with rigid-body biomechanics to predict human motion 100–500ms ahead for Human-Robot Collaboration.
- Integrating biomechanical joint dynamics into the training objective through physics-consistency loss, constraining predictions to physically plausible human motion under noisy or occluded observations.
- Benchmarking against LSTM/Transformer baselines using MPJPE, bone-length consistency, joint-limit violations, data efficiency, and occlusion robustness across 100ms, 200ms, and 500ms prediction horizons.

EXTRACURRICULAR ACHIEVEMENTS

Lead, Operations Department | *Vellore Institute of Technology — Android Club* April 2024 - April 2026

- Directed a cross-functional team of 80+ members across 25+ technical events for 1,000+ participants, owning scheduling, logistics, stakeholder communication, and budgets.
- Redesigned event-planning and coordination workflows, cutting planning time by 20% while maintaining on-schedule delivery across all events.

CERTIFICATIONS

- **NPTEL – Introduction to Machine Learning:** covered core ML algorithms, model evaluation, and supervised/unsupervised learning fundamentals.
- **NPTEL – Marketing Analytics:** covered data-driven customer segmentation, campaign measurement, and marketing decision-making.
- **Google / Coursera – Bits and Bytes of Computer Networking:** covered networking fundamentals, protocols, and the OSI model.
- **Adobe – UI/UX:** covered core principles of interface and experience design, including wireframing and prototyping.
- **ServiceNow – System Administrator:** covered platform administration, workflow configuration, and IT service management.